

THORSBY AGCM, AB, Canada

WMO: 715040

Lat: 53.2172N

Lon: 113.8947W

Elev: 2474

StdP: 13.43

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.2	-17.4	-29.8	1.1	-23.0	-23.8	1.6	-17.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	81.7	61.9	78.9	61.1	75.9	59.9	66.0	76.6	63.9	74.5	62.0	72.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	91.6	71.6	59.7	83.9	69.2	57.7	77.8	66.9	32.2	76.4	30.4	74.5	29.0	72.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-32.1	88.3	7.0	4.0	-37.2	91.2	-41.2	93.5	-45.2	95.8	-50.2	98.7
			-32.4	69.7	6.9	3.1	-37.3	71.9	-41.4	73.7	-45.3	75.5	-50.3	77.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	15.9	13.8	24.0	37.7	50.8	57.6	61.6	60.0	51.7	39.4	24.8	12.7
	DBStd	20.98	15.53	15.50	14.08	9.92	7.91	5.25	4.79	5.47	8.08	8.39	12.99	14.42
	HDD50	5676	1058	1014	805	381	86	6	0	2	73	339	756	1156
	HDD65	10032	1523	1434	1270	820	441	229	125	170	401	792	1206	1621
	CDD50	1169	0	0	0	11	113	233	361	313	125	12	1	0
	CDD65	47	0	0	0	0	2	6	21	16	2	0	0	0
	CDH74	1048	0	0	0	4	101	122	360	304	153	4	0	0
	CDH80	173	0	0	0	0	10	12	61	57	33	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	19.1	0.8	0.6	0.6	0.9	2.0	3.3	3.8	3.0	1.9	0.9	0.7	0.7
	PrecMax	23.4	2.2	1.6	1.5	2.2	6.3	7.7	9.3	7.5	5.7	2.4	1.9	1.6
	PrecMin	14.9	0.0	0.0	0.1	0.0	0.3	0.6	1.3	0.2	0.2	0.1	0.1	0.2
	PrecStd	2.3	0.6	0.4	0.4	0.6	1.2	1.7	1.9	1.7	1.1	0.6	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	46.5	54.8	73.0	81.5	81.6	85.8	85.4	84.2	71.4	59.3	44.5
		MCWB	37.4	38.1	43.6	51.4	55.1	62.0	63.6	63.7	60.7	53.6	45.7	36.9
	2%	DB	43.3	41.2	48.6	65.6	77.0	77.5	81.2	80.9	78.3	63.1	50.3	40.2
		MCWB	36.2	34.4	39.2	47.6	53.5	59.6	64.2	62.7	58.1	48.9	41.0	33.9
	5%	DB	40.3	37.9	43.8	58.8	72.3	74.2	78.2	77.3	73.2	58.2	44.1	36.8
		MCWB	34.5	32.7	36.7	43.5	52.2	58.0	63.3	62.7	56.7	46.0	36.9	31.2
	10%	DB	37.1	35.0	39.9	53.8	67.7	70.9	74.7	73.7	67.9	53.5	40.5	32.9
		MCWB	32.2	30.7	34.0	41.7	50.2	56.2	61.7	61.1	54.2	43.6	34.9	28.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.7	44.3	51.8	57.3	65.1	69.9	68.6	62.0	53.8	46.4	37.5
		MCDB	45.1	45.8	53.6	72.1	75.3	77.5	80.4	77.5	77.8	69.2	58.1	43.8
	2%	WB	36.4	35.1	39.9	47.9	55.3	61.7	66.6	65.8	59.5	50.1	41.7	34.4
		MCDB	42.5	40.3	47.5	64.3	71.7	72.9	76.8	76.4	75.8	60.6	48.9	39.6
	5%	WB	34.6	33.0	37.0	44.9	53.5	59.6	64.6	63.6	57.1	47.0	38.1	31.7
		MCDB	39.8	37.5	43.5	57.5	69.5	70.1	75.1	74.3	71.3	56.3	43.5	36.3
	10%	WB	32.0	30.8	34.4	42.1	51.6	57.8	62.7	61.8	55.0	44.5	35.3	28.6
		MCDB	36.3	35.0	39.3	52.2	65.8	68.2	72.7	72.6	66.4	51.7	39.8	32.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	19.2	18.6	20.1	23.7	21.1	21.9	22.9	23.7	20.1	17.0	16.9
		MCDBR	17.9	20.3	21.7	30.3	31.1	27.3	28.1	29.3	34.5	28.7	21.3	18.6
		MCWBR	13.5	15.1	14.0	15.6	13.8	12.9	14.3	14.7	16.8	16.4	13.8	14.3
		MCDBR	18.1	20.0	21.0	28.8	27.3	24.0	25.4	26.0	31.7	26.6	20.1	19.2
	5% WB	MCWBR	13.8	15.0	13.8	15.4	12.9	12.6	14.0	14.5	16.1	15.6	13.4	15.5
(40) Clear-Sky Solar Irradiance	taub		0.254	0.263	0.304	0.328	0.342	0.349	0.353	0.360	0.308	0.283	0.254	0.240
	taud		2.321	2.379	2.363	2.384	2.390	2.387	2.421	2.356	2.494	2.547	2.464	2.319
	Ebn at Noon		236	274	282	288	288	285	282	273	278	265	240	220
	Edh at Noon		18	23	30	33	35	36	34	34	26	20	16	15
(44) All-Sky Solar Radiation	RadAvg		316	641	1112	1479	1794	1897	1956	1612	1118	650	322	232
	RadStd		29	49	78	137	126	126	106	110	116	73	33	21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page